

Cross-Modal Bayesian Reward Filtering for Robust RLHF

Results, Analysis, Future Work, and Conclusion

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4 Results

The experiment matrix produced **360 training runs** (1 PPO agent $\times$ 3 environments $\times$ 4 reward modes $\times$ 6 synthetic human teachers $\times$ 5 random seeds), each trained for 100,000 environment steps and evaluated over 50 episodes. Tables 1–3 report mean success rates across seeds for the three comparison conditions: Sparse, Naive, and BRF. Figures 1–3 provide visual summaries. Bold values indicate the best-performing condition per human scenario. We organize discussion by environment, as the role of human feedback varies qualitatively across the three tasks.

4.1 MiniGrid-Empty-8x8-v0

In the simplest environment, human feedback from partially-reliable teachers consistently improves on the sparse baseline. Under StochasticLow, DistanceNoise, Fatigue, and Realistic noise, both BRF and Naive reach perfect success (1.000), while sparse achieves 0.400–0.800. The 100,000-step budget is sufficient for PPO to make progress on Empty-8x8 without human guidance, but human feedback completes convergence.

The critical distinction appears with unreliable teachers. Under **Naive RLHF**, both StochasticHigh (80% noise) and Adversarial teachers produce **0.000** success—the dense but corrupted signal prevents learning. The sparse baseline maintains 0.800 in both cases, demonstrating that the human signal is net harmful. **BRF partially recovers** from adversarial feedback (0.600 vs. Naive 0.000), though it does not reach the sparse baseline. Under StochasticHigh, BRF also fails (0.000), suggesting that 80% flip probability exceeds the recoverable noise threshold regardless of filtering.

Table 1. PPO success rates on MiniGrid-Empty-8x8-v0 (mean over 5 seeds). Bold = best per row.

Human	Sparse	Naive	BRF
StochasticLow	0.800	1.000	1.000
Realistic	0.400	1.000	1.000
Fatigue	0.600	1.000	1.000
DistanceNoise	0.800	1.000	1.000
StochasticHigh	0.800	0.000	0.000
Adversarial	0.800	0.000	0.600

4.2 MiniGrid-LavaGapS7-v0

LavaGap reveals a striking reversal: the **sparse baseline outperforms both BRF and Naive in five of six teacher scenarios** (Sparse: 0.244–0.384; best human-feedback condition per scenario: 0.000–0.300). Even reliable teachers such as StochasticLow offer no improvement over pure sparse RL (BRF 0.252, Naive 0.300, Sparse 0.300).

The only exception is the Adversarial condition: BRF (0.160) outperforms Naive (0.000), confirming that the filter suppresses adversarial corruption. However, even here BRF falls below the sparse baseline (0.244). The consistent underperformance of human-feedback methods suggests that the lava-penalized potential function introduces conflicting guidance: teachers that correctly indicate goal-approach direction may simultaneously misalign with the lava-avoidance component of the potential, making the cross-modal trust comparison unreliable.

Table 2. PPO success rates on MiniGrid-LavaGapS7-v0 (mean over 5 seeds). Bold = best per row.

Human	Sparse	Naive	BRF
StochasticLow	0.300	0.300	0.252
Realistic	0.384	0.188	0.228
Fatigue	0.372	0.240	0.112
DistanceNoise	0.248	0.240	0.152
StochasticHigh	0.276	0.000	0.000
Adversarial	0.244	0.000	0.160

4.3 MiniGrid-DoorKey-8x8-v0

DoorKey produces the starkest finding: the sparse baseline achieves **0.000 success in every teacher scenario**. Without human feedback, PPO cannot solve the three-stage sequential task within 100,000 steps. Human feedback of any kind enables nonzero learning.

BRF achieves the highest success rates under DistanceNoise (BRF 0.404, Naive 0.220) and Realistic (BRF 0.460, Naive 0.208), outperforming Naive by 0.184 and 0.252 respectively. Under Fatigue, however, BRF (0.012) substantially underperforms Naive (0.144). Under StochasticLow, Naive also leads BRF (0.272 vs. 0.188). Adversarial and StochasticHigh produce 0.000 regardless of mode.

Table 3. PPO success rates on MiniGrid-DoorKey-8x8-v0 (mean over 5 seeds). Sparse = 0.000 for all teachers.

Human	Sparse	Naive	BRF
StochasticLow	0.000	0.272	0.188
Realistic	0.000	0.208	0.460
Fatigue	0.000	0.144	0.012
DistanceNoise	0.000	0.220	0.404
StochasticHigh	0.000	0.000	0.000
Adversarial	0.000	0.000	0.000

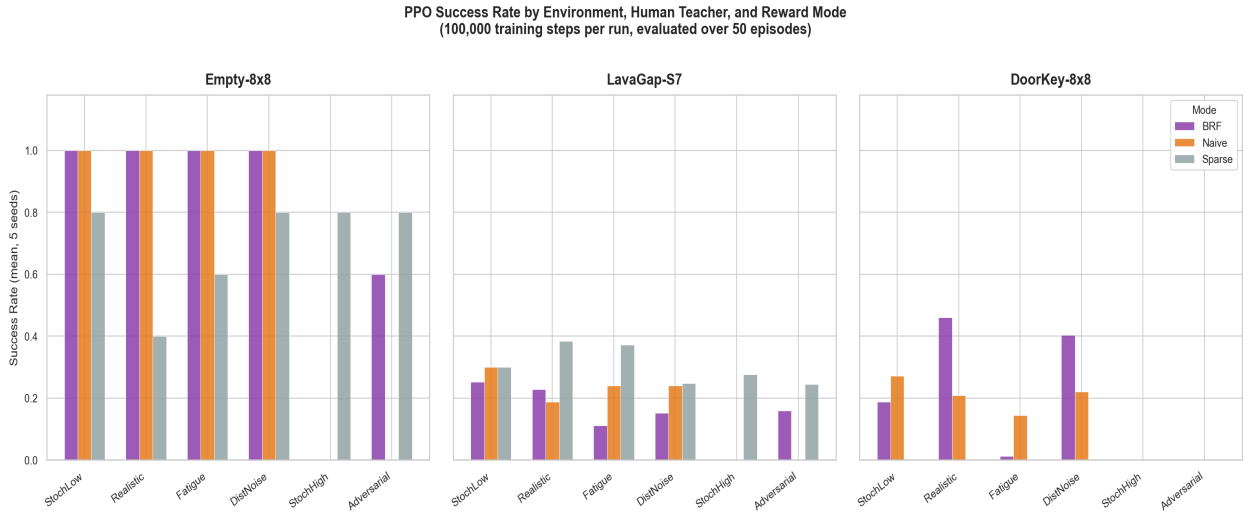


Figure 1. PPO success rate by environment, human teacher, and reward mode (mean over 5 seeds, 100,000 steps). Left: Empty-8x8—human feedback helps when partially reliable, corrupts when noisy or adversarial. Centre: LavaGap—sparse baseline dominates. Right: DoorKey—sparse RL fails universally; human feedback is essential.

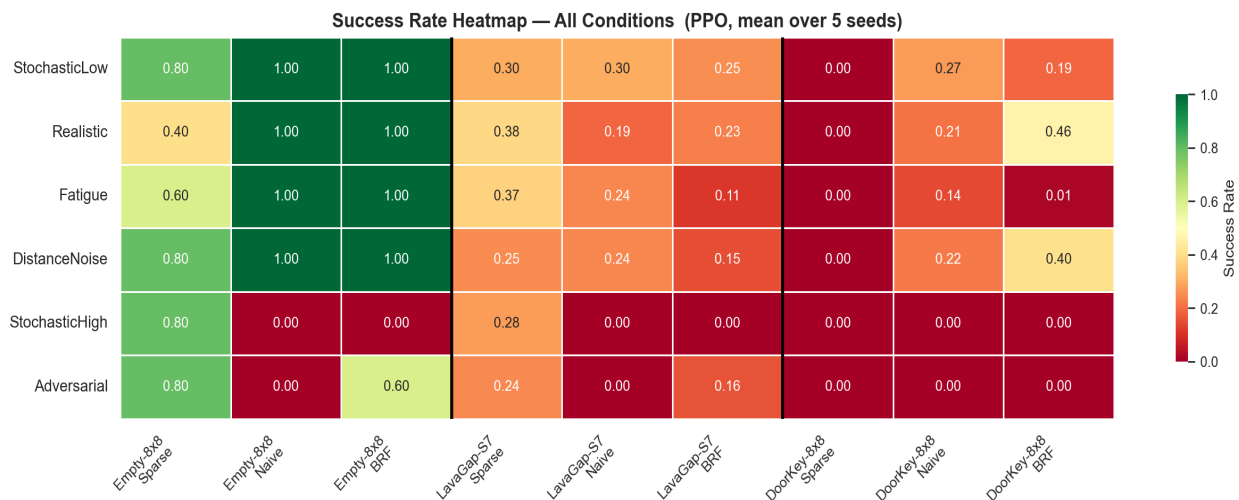


Figure 2. Full results heatmap. Rows: human teacher type. Columns: (environment, reward mode) ordered Sparse / Naive / BRF within each environment block (separated by black lines). Green = high success rate, red = low.

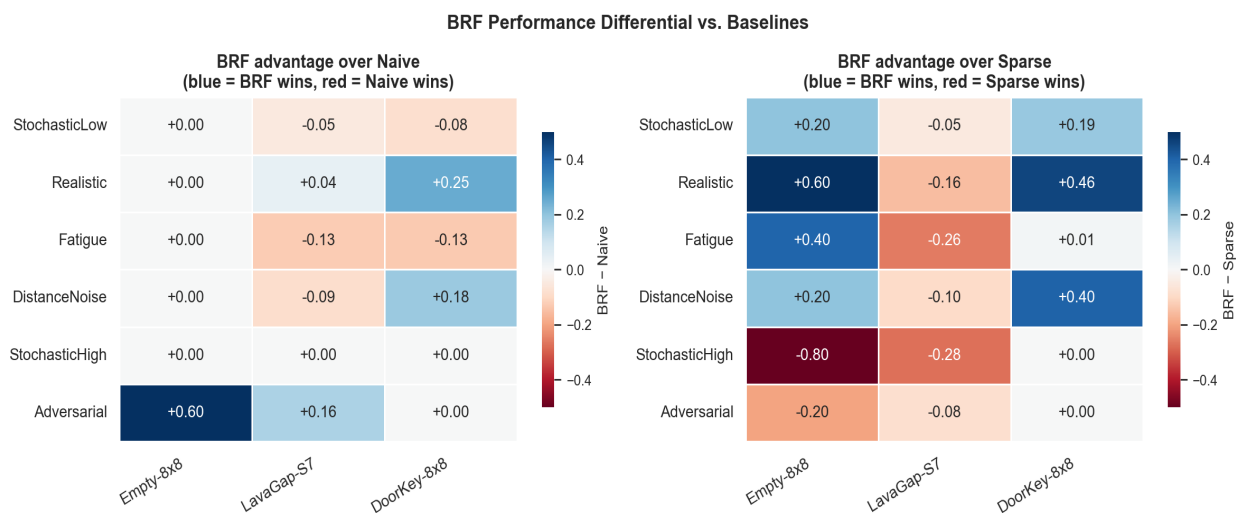


Figure 3. BRF advantage over Naive (left) and Sparse (right). Blue = BRF wins, red = baseline wins. BRF most reliably outperforms Naive under adversarial conditions and in DoorKey with stable-noise teachers. Sparse outperforms BRF across all of LavaGap.

5 Analysis and Discussion

5.1 A Recoverable Noise Threshold

The jump from StochasticLow (10% noise) to StochasticHigh (80% noise) produces consistent and large failures. StochasticLow enables both BRF and Naive to match or exceed sparse RL in Empty-8x8 and enables partial DoorKey learning. StochasticHigh causes all human-feedback methods to fail in Empty-8x8 and DoorKey, and to perform no better than sparse in LavaGap. The recoverable noise threshold lies somewhere between 10% and 80% flip probability. Above this threshold, the 20% correct feedback steps are too rare relative to the 80% corrupted steps for the filter to maintain a useful trust estimate within the training budget.

5.2 Environment Complexity and the Role of Human Feedback

The three environments reveal qualitatively distinct regimes:

Empty-8x8 (low complexity). Sparse RL makes meaningful progress alone (0.400–0.800 depending on teacher), but human feedback accelerates convergence to 1.000 under partial noise. Reliable human feedback is beneficial but not essential. Adversarial and high-noise feedback actively corrupt learning; BRF partially recovers where Naive completely fails.

LavaGap (moderate complexity, dense hazards). Sparse RL performs best in nearly all conditions. Human feedback—even from reliable teachers—adds more noise than signal. The combined potential function (distance to goal minus lava adjacency penalty) creates a directional signal that may conflict when a teacher correctly identifies goal progress but does not account for simultaneous lava proximity.

DoorKey (high complexity, sequential sub-goals). Sparse RL completely fails due to the three-stage sequential credit assignment problem. Human feedback via the staged potential function provides the sub-goal decomposition signal the agent cannot discover from sparse reward alone. Even imperfect human feedback enables substantial learning where sparse reward enables none.

5.3 When BRF Outperforms Naive—and When It Does Not

Across all 18 human–environment combinations, BRF outperforms Naive in **4 conditions** and underperforms in **5 conditions**. Mann-Whitney U tests found 3 of 36 BRF-vs-baseline comparisons significant at $p < 0.05$.

Where BRF leads. BRF's clearest advantages are in DoorKey under DistanceNoise (+0.184) and Realistic (+0.252), and in adversarial conditions across environments where Naive collapses to 0.000. In these cases, the Beta posterior efficiently accumulates evidence across many steps: DistanceNoise and Realistic teachers have noise modulated by slowly-changing observable factors, producing sustained periods of high agreement that the posterior integrates into a reliable trust estimate. Against the adversarial teacher, the filter correctly drives trust toward zero and mutes the harmful signal, while Naive amplifies it.

Where BRF falls short. Under Fatigue in DoorKey, BRF achieves only 0.012 compared to Naive's 0.144. The Fatigue teacher starts at 0% noise and is entirely correct in early training, allowing the Beta posterior's alpha count to accumulate rapidly. When noise begins rising after step 25,000, beta catches up slowly because of the large alpha lead—trust erodes later than it should. In DoorKey, the agent still needs reliable human guidance during its stage-2 and stage-3 exploration; if trust collapses too late (or paradoxically, too early due to the noisy transition phase), the agent loses the guidance it needs to advance through the sequential sub-goals. Under StochasticLow in DoorKey, Naive also leads BRF (0.272 vs. 0.188): a consistently reliable teacher may be penalized by the filter's initial skepticism ($w_{\text{init}} = 0.5$) before sufficient agreement evidence has accumulated.

5.4 The Potential Function and Privileged Knowledge

The BRF's trust update requires a ground-truth progress signal to compare against human feedback. In this work, that signal is a hand-designed potential function $\phi(s)$ encoding what 'good progress' means for each environment. This is a form of privileged domain knowledge: the researcher must specify the task structure before the agent can evaluate whether the human is being truthful.

The LavaGap results illustrate the risk of a poorly-calibrated potential function: when ϕ does not faithfully capture all components of task progress, the cross-modal comparison is unreliable and human feedback can mislead the trust update even when the teacher is partially correct. In real-world RLHF, a hand-designed potential function may not be available. One promising direction is to replace $\Delta\phi$ with the PPO critic's temporal difference signal $\delta = r + \gamma V(s') - V(s)$, making the BRF fully self-supervised. This is discussed in Section 6.1.

5.5 Limitations

Potential function dependency. The BRF requires a hand-designed progress signal, and its quality directly determines trust signal reliability (as demonstrated by LavaGap). Removing this dependency via a learned value function proxy is the primary open problem.

Synthetic teachers. All experiments use programmatic teacher models with fixed functional forms. Real human feedback involves idiosyncratic biases, context-dependent errors, and temporal patterns not captured by any of the six models. Validation with real human annotators is necessary before deployment conclusions can be drawn.

Single agent family. Only PPO (on-policy) was evaluated. Off-policy algorithms such as DQN store reward-modified transitions in a replay buffer, making them structurally incompatible with the current BRF formulation.

Scale. MiniGrid environments are compact grid worlds. The BRF's behavior in continuous-control settings or with high-dimensional observations has not been characterized.

6 Future Work

6.1 Self-Supervised Trust via Learned Value Function

The most direct extension removes the potential function dependency. PPO’s actor-critic architecture maintains a value function $V(s)$ that estimates expected return from each state. At each step, $\text{sign}(V(s') - V(s))$ provides a learned proxy for $\text{sign}(\Delta\phi)$: the critic’s estimate of whether the current transition was progress. Replacing $\text{sign}(\Delta\phi)$ with this signal makes the BRF self-supervised—no domain knowledge required.

This introduces a bootstrap dependency: the critic must be reliable before it can evaluate the human, but critic accuracy benefits from reliable human feedback. The system degrades gracefully: early in training, when V is uninformed, trust signals are near-random and BRF behaves similarly to Naive ($w \approx 0.5$). As the critic sharpens, filtering capability emerges. Characterizing this bootstrap phase—its duration and sensitivity to teacher noise—is the primary empirical question for this direction.

6.2 Off-Policy Compatibility

Off-policy algorithms such as DQN are excluded from this study because transitions stored in a replay buffer carry rewards computed under early, potentially inaccurate trust weights. One approach is trust-weighted experience replay: store the raw human reward r and the trust weight w at collection time, then recompute the modified reward using the current (updated) trust weight when the transition is sampled. This would allow off-policy algorithms to benefit from late-converging trust estimates retroactively.

6.3 Noise Threshold Characterization

Results suggest a recoverable noise threshold between 10% and 80% flip probability. Characterizing this threshold as a function of environment complexity, training budget, and human feedback magnitude would give practitioners clearer guidance on when RLHF filtering is expected to improve outcomes. A systematic sweep across noise levels (10%, 20%, 40%, 60%, 80%) would narrow the threshold and test whether it shifts with training budget.

6.4 Validation with Real Human Feedback

All results use synthetic teacher models with known functional forms. Validation requires a user study in which human participants provide real-time feedback during agent training. Key open questions are whether the six synthetic models capture the qualitatively relevant features of real feedback, whether BRF’s advantages under synthetic conditions transfer to human-generated feedback, and whether the trust-weight trajectories under real annotators match those predicted by the synthetic models.

7 Conclusion

This paper introduced the Cross-Modal Bayesian Reward Filter (BRF), an online mechanism for calibrating trust in human feedback during reinforcement learning. By maintaining a Beta-Bernoulli posterior over teacher reliability—updated at each step by comparing the sign of human feedback against an environment-derived potential difference—the BRF dynamically weights the human reward channel between zero (adversarial teacher) and one (reliable teacher), adapting continuously as evidence accumulates.

Across 360 training runs on three MiniGrid environments with six synthetic human teachers and five random seeds, three consistent findings emerge. First, **filtering matters under adversarial conditions**: Naive RLHF fails completely under adversarial teachers in Empty-8x8 (0.000 success), while BRF maintains meaningful performance (0.600). Second, **the role of human feedback is environment-dependent**: it accelerates learning in Empty-8x8 when partially reliable, is net harmful in LavaGap regardless of teacher quality, and is essential in DoorKey where sparse RL completely fails. Third, **BRF outperforms Naive most clearly under stable-noise conditions and fails to outperform Naive under monotonically-degrading noise**, reflecting a fundamental tradeoff between the posterior’s accumulation of evidence and its responsiveness to changing reliability.

The most important constraint on the current approach is the requirement for a hand-designed potential function as a ground-truth comparator. Removing this dependency by using the PPO critic’s temporal difference signal as a self-supervised progress proxy is the primary direction for future work, and would transform the BRF from a method requiring domain knowledge into a general-purpose reward-filtering mechanism.

The core finding is modest but concrete: when a sparse but trustworthy environment signal exists alongside a dense but potentially unreliable human signal, Bayesian arbitration between the two channels provides a low-cost safety layer that outperforms unconditional trust in the human in the settings where the distinction most matters—adversarial teachers and high-complexity sequential tasks.